

Script-Controlled ACL – Restrict Record Access

Based on field value

Project overview

Milestone 1: Creation of users and roles

User Creation:

- 1. Log in** to the ServiceNow instance with admin access.
- 2. Create a test user:**

- Navigate to User Administration → Users
- Click New
- **User ID:** EEE User
- **First name:** EEE
- **Last name:** User
- **Email:** eeeuser@gmail.com
- Save the record

User ID	Name	Email	Active	Created	Updated
eeeuser@gmail.com	EEE User	eeeuser@gmail.com	true	2026-09-22 22:24:50	2026-09-22 22:24:50
efren.baucher	Efren Baucher	efren.baucher@example.com	true	2012-02-17 19:04:53	2026-09-21 19:23:53
elaine.renzi	Elaine Renzi	elaine.renzi@example.com	true	2012-02-17 19:04:52	2026-09-21 19:23:23
eldon.sutch	Eldon Sutch	eldon.sutch@example.com	true	2012-02-17 19:04:53	2026-09-21 19:25:45
eli.bettner	Eli Bettner	eli.bettner@example.com	true	2012-02-17 19:04:51	2026-09-21 19:23:58
elisa.gracely	Elisa Gracely	elisa.gracely@example.com	true	2012-02-17 19:04:52	2026-09-21 19:23:20
eliseo.wice	Eliseo Wice	eliseo.wice@example.com	true	2012-02-17 19:04:53	2026-09-21 19:23:28
ella.pahnke	Ella Pahnke	ella.pahnke@example.com	true	2012-02-17 19:04:49	2026-09-21 19:25:07
elmo.dagenais	Elmo Dagenais	elmo.dagenais@example.com	true	2012-02-17 19:04:53	2026-09-21 19:25:21
elmo.gabouer	Elmo Gabouer	elmo.gabouer@example.com	true	2012-02-17 19:04:53	2026-09-21 19:23:08
elvira.blumenthal	Elvira Blumenthal	elvira.blumenthal@example.com	true	2012-02-17 19:04:52	2026-09-21 19:24:52
emery.reek	Emery Reek	emery.reek@example.com	true	2012-02-17 19:04:49	2026-09-21 19:23:30
emilia.oxley	Emilia Oxley	emilia.oxley@example.com	true	2012-02-17 19:04:49	2026-09-21 19:24:59
emilio.lampkin	Emilio Lampkin	emilio.lampkin@example.com	true	2012-02-17 19:04:49	2026-09-21 19:25:43
enrique.oroark	Enrique Oroark	enrique.oroark@example.com	true	2012-02-17 19:04:50	2026-09-21 19:23:40

Role Creation

- Navigate to User Administration → Roles
- Click New
- Name: Enter role name as bb1
- Description: Brief role purpose
- Click Submit
- Create other roles also like bb2,bb3,bb4.(follow the same procedure)
- Assign all the roles to the EEE User

Name	Description	Elevated privilege
bb1		false
bb2		false
bb3		false
bb4		false
bsm_legacy	Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View	false
bsm_legacy_admin	Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View	false
business_calendar_admin	Access to configure and manage plugins and features related to business calendars.	false
business_process_admin		false
business_process_manager		false
business_process_user		false
business_relationship_manager	Person(s) responsible for both business strategy and business value in an effort to advance business objectives.	false
business_rule_admin	Can manage Business Rules	false
business_stakeholder	Use the Business Stakeholder role to provide authorized users to approve, view/read records across the organization and view reports. Note there is a fee associated with this role. Do not assign it to users without confirming your organization has the appropriate entitlement. Please review the documentation for additional details.	false
caching_admin	Role for monitoring and configuring caching framework related tables	false

Milestone 2:

Tables Creation

1. In the Application Navigator, search for System Definition → Tables.
2. Click New and create a custom table with the following details:
 - Label: Institution Details
 - Name: u_institution_details
 - Extends for : false
3. Click Submit.
4. Add the following fields to the u_institution_details table:

- Student Roll Number– Auto Number
- Student Name – Reference- User
- Faculty Name – Reference- User
- Branch – Choice (ECE, EEE, CSE)
- Email– String
- Phone Number– String
- Description– Multi String

5. Save the Table.

Label	Name	Extends table	Extensible	Updated
Search	Search	Search	Search	Search
Institution Details	u_institution_details	(empty)	false	2026-09-22 22:39:52
Institution Details	u_st_u_institution_details	(empty)	false	2026-09-22 22:43:06
Validator Run Summary	validator_run_summary	(empty)	false	2026-06-12 14:38:11
Variant	variant	(empty)	false	2026-06-12 14:03:08
Variables	var_dictionary	Dictionary Entry	true	2026-06-12 13:58:20
Menu Item	va_branding_contact_menu	Application File	false	2026-06-12 14:00:49
VMware Datastore HostMount	vcenter_datastore_hostmount	(empty)	false	2026-06-12 14:23:06
vCenter Event	vcenter_event	(empty)	false	2026-06-12 14:23:01
vCenter Event vCenter Extension Context	vcctr_evt_vcctr_ext_ctx_m2m	(empty)	false	2026-06-12 14:23:16
Vendor Type	vendor_type	(empty)	false	2026-06-12 13:57:02
Map source	viz_map_source	(empty)	false	2026-06-12 15:00:48
VMware Credentials	vmware_credentials	Credentials	false	2026-06-12 14:23:16
VMware Folder Type	vmware_folder_type	Application File	false	2026-06-12 14:23:07
vCenter Folder Type	vmware_vcenter_folder_type_m2m	(empty)	false	2026-06-12 14:23:16
Visual Task Board	vtb_board	(empty)	false	2026-06-12 14:01:59

6. Create multiple records in the Student Records table with different branch values (ECE, EEE, CSE).

dev428056.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D5a1979bc73a7c310638cfb204ab8b7c0%26sysparm_record_targ...

servicenow All Favorites History Workspaces Table - Institution Details Search

Table Institution Details Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
×	Email	String	(empty)	40		false
×	Student Roll Number	String	(empty)	40		false
×	Branch	Choice	(empty)	40		false
×	Phone Number	String	(empty)	40		false
	Updates	Integer	(empty)	40		false
×	Faculty Name	Reference	User	32		false
	Sys ID	Sys ID (GUID)	(empty)	32		false
×	Description	String	(empty)	40		false
	Created by	String	(empty)	40		false
	Created	Date/Time	(empty)	40		false
	Updated by	String	(empty)	40		false
	Updated	Date/Time	(empty)	40		false
×	Student Name	Reference	User	32		false
+	Insert a new row...					

Delete Update Delete All Records

Related Links
orm Builder

Milestone 3:

Creation of Access Control Lists(Read ACL)

1. In the Application Navigator, search for System Security → Access Control (ACL).

2. Elevate “Security_admin” role

3. Click New to create a new ACL.

4. Enter the following details:

○ Type: record

○ Operation: read

○ Name: u_institution_details

○ Active: true

○ Advanced: true

5. In the Requires role related list, add:

○ bb1 (custom role)

6. Add Data condition as Branch is EEE

7. In the Script field, add the following script:

```
(function () {  
    // Allow admin users full access  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
    // Allow only EEE branch users to see EEE records  
    if (gs.hasRole('bb1')){  
        return true;  
    }  
    // Deny access for all others  
    return false;  
})();
```

8. Click Submit to save the ACL.

dev428056.service-now.com/now/nav/ui/classic/params/target/sys_user_role_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3DGOTOname%253e%253...

servicenow

All Favorites History Workspaces Admin

Roles

Search

Roles

Name

Search

ul > Name >= bb1

Name

Description

Search

Search

bb1

bb2

bb3

bb4

bsm_legacy

Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View

bsm_legacy_admin

Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View

business_calendar_admin

Access to configure and manage plugins and features related to business calendars.

business_process_admin

business_process_manager

business_process_user

business_relationship_manager

Person(s) responsible for both business strategy and business value in an effort to advance business objectives.

business_rule_admin

Can manage Business Rules

business_stakeholder

Use the Business Stakeholder role to provide authorized users to approve, view/read records across the organization and view reports. Note there is a fee associated with this role. Do not assign it to users without confirming your organization has the appropriate entitlement. Please review the documentation for additional details.

caching_admin

Role for monitoring and configuring caching framework related tables

false

false

false

false

false

false

1 to 20 of 700

System Administrator

System Administrator

ServiceNow

Profile

Preferences

Keyboard shortcuts

Impersonate user

Elevate role

Printer friendly version

Log out

dev428056.service-now.com/now/nav/ui/classic/params/target/sys_security_acl_list.do%3Fsysparm_userpref_module%3D4ec206740a0aa700b850e0337b2f00...

servicenow

All Favorites History Workspaces Admin

Access Controls

Search

Access Controls

Name

Search

Name

Decision Type

Operation

Type

Active

Updated by

\$allappsmgmt

Allow If

read

ui_page

true

admin

\$atf_page_inspector

Allow

admin

\$conversation-builder

Allow

admin

\$mycompanyappsmgmt

Allow

admin

\$products

Allow

admin

\$pwd_reset_serviceDesk

Allow

admin

\$sa_service_model_json

Allow

admin

\$sla_timeline

Allow

admin

\$sn-va-web-client-app

Allow

admin

\$sn_global_search_results

Allow

admin

\$spd

Allow

admin

\$spm_availability_calculation_setting

read

ui_page

true

admin

\$studio

Allow If

read

ui_page

true

admin

\$upgrade_client

Allow If

read

ui_page

true

admin

\$upgrade_monitor

Allow If

read

ui_page

true

admin

\$upgrade_preview

Allow If

read

ui_page

true

admin

2020-06-15 10:43:57

2016-01-19 12:53:23

2024-02-18 06:04:39

2016-01-09 16:21:38

2025-11-10 16:44:04

2025-09-29 22:40:08

2025-09-23 03:04:40

Elevate role

Elevate a role by adding privileges, which end when you log out. [Learn more](#)

AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel Update

1 to 20 of 44,195

System Administrator

System Administrator

ServiceNow

Profile

Preferences

Keyboard shortcuts

Impersonate user

Elevate role

Printer friendly version

Log out

dev428056.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D33d18630732bc310638cfb204ab8b755%26sysparm_record_tar...

servicenow

All Favorites History Workspaces

Access Control - u_institution_details

Search

Access Control

u_institution_details

Update

Delete

↑

↓

* Type

record

Application

Global

* Operation

read

Active

☒

Admin overrides

☒

Advanced

☒

Decision Type

Allow If

Protection policy

-- None --

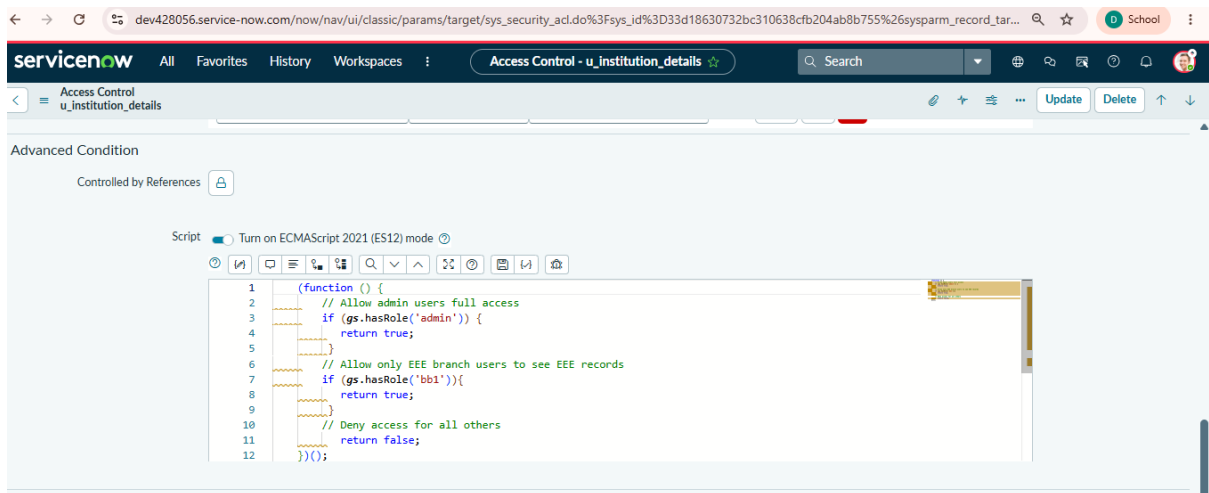
* Name

Institution Details [u_institution_details]

-- None --

Description

Allow read for records in u_institution_details, for users with role bb 1, and if the ACL script returns true, and if the ACL condition (u_branch=EEE*EQ) evaluates to true.



9. Impersonate EEE users and open the Student Records list.

10. Copy the table name u_institution_details.list/LIST and paste in the All menu.

11. Impersonate Admin and open the Student Records list.

Verify :

- User with bb1 role can view only EEE branch records.
- User without the role cannot view any records
- Admin user can view all records regardless of branch.

Milestone 4:

Creation of Access Control Lists(Create ACL)

1. Click New to create a new ACL(Create)

● User with bb1 &bb2 roles can view only EEE branch records b New Button.

Milestone 5:

Creation of Access Control Lists(Write ACL)

12. Click New to create a new ACL(Write)

13. Enter the following details:

○ Type: record

○ Operation: Write

○ Name: u_institution_details

○ Active: true

14. In the Requires role related list, add:

○ bb3 (custom role)

15. No need to add any Data condition

16. Save the ACL Record

The screenshot shows the ServiceNow 'Access Control' form for the record 'u_institution_details'. The form is titled 'Access Control - u_institution_details' and includes a search bar and navigation links. The form fields are as follows:

- * Type: record
- * Operation: write (highlighted with a green border)
- Admin overrides: ☒
- Decision Type: Allow If
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details] (dropdown menu)
- Description: Allow write for records in u_institution_details, for users with role bb3.
- Application: Global
- Active: ☒
- Advanced: ☐

At the bottom of the form, there is a section labeled 'Conditions' with a dropdown arrow.

Verify :

● User with bb1 &bb2 &bb3 roles can view and edit the EEE branch records along with New Button.

Milestone 6:

Creation of Access Control Lists(Delete ACL)

17. Click New to create a new ACL(Delete)

18. Enter the following details:

○ **Type:** record

○ **Operation:** Delete

○ **Name:** u_institution_details

○ **Active:** true

19. In the Requires role related list, add:

○ bb4 (custom role)

20. No need to add any Data condition

21. Save the ACL Record .

The screenshot displays the ServiceNow 'Access Control' console for creating a new ACL record. The breadcrumb trail shows 'Access Control - u_institution_details'. The form fields are as follows:

- Type:** record
- Operation:** delete
- Application:** Global
- Active:** ☒
- Advanced:** ☐
- Admin overrides:** ☒
- Decision Type:** Allow If
- Protection policy:** -- None --
- Name:** Institution Details [u_institution_details] (selected from a dropdown)
- Description:** Allow delete for records in u_institution_details, for users with role bb4.

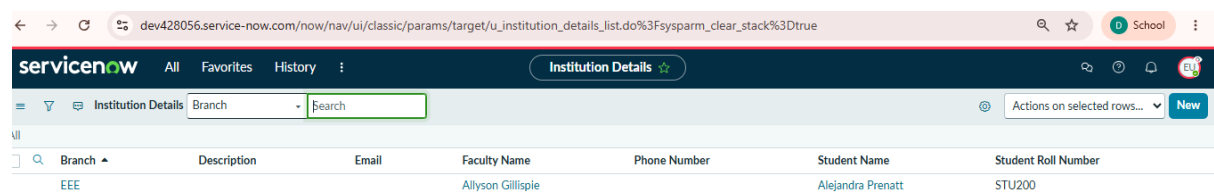
At the bottom, there is a 'Conditions' section with a downward arrow.

Verify :

- Users with bb1 &bb2 &bb3&bb4 roles can see, create and edit along with delete the EEE branch records.

Milestone 7:

Conclusion



Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Allyson Gillispie		Alejandra Prenatt	STU200